

Professional Summary:

- AI/ML Engineer with 9+ years of experience in designing, developing, and deploying scalable machine learning and deep learning solutions to solve complex business problems.
- Expertise in building end-to-end ML pipelines, including data ingestion, preprocessing, feature engineering, model development, evaluation, deployment, and monitoring in production environments.
- Strong background in deep learning architectures such as CNNs, RNNs, and Transformer-based models for NLP, computer vision, and time-series applications.
- Experienced in developing Generative AI applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and prompt engineering techniques.
- Proficient in implementing MLOps practices including CI/CD pipelines, model versioning, experiment tracking, and automated retraining for scalable and reliable AI systems.
- Skilled in deploying machine learning models as RESTful APIs using Flask and FastAPI for real-time inference and enterprise integration.
- Strong expertise in data preprocessing, feature engineering, exploratory data analysis, and statistical modeling to improve model performance and insights.
- Hands-on experience working with large-scale structured and unstructured datasets using Python, SQL, and distributed data processing frameworks.
- Proficient in optimizing model performance using hyperparameter tuning, cross-validation, ensemble learning, and regularization techniques.
- Familiar with cloud platforms such as AWS, Azure, and GCP for scalable model training, deployment, and infrastructure management.
- Adept at translating business requirements into AI-driven solutions and delivering measurable impact through data-driven strategies.
- Strong experience in implementing end-to-end MLOps frameworks for continuous integration, deployment, monitoring, and lifecycle management of ML models.
- Hands-on experience with AWS SageMaker pipelines and Databricks for scalable ML workflows, experiment tracking, distributed model training, and enterprise data processing.
- Adept at developing robust data pipelines and feature engineering strategies to enhance model accuracy, scalability, and reliability.

Technical Skills:

Programming	Python, SQL
Machine Learning	Scikit-learn, XGBoost, LightGBM, Ensemble Methods
Deep Learning	TensorFlow, PyTorch, Keras
NLP & GenAI	Transformers, Hugging Face, LLMs, RAG, Prompt Engineering
Computer Vision	CNNs, OpenCV
Data Processing	Pandas, NumPy, Apache Spark
MLOps & Tools	MLflow, Docker, Kubernetes, Airflow, CI/CD, AWS SageMaker, Databricks
Cloud Platforms	AWS / Azure / GCP
Databases	MySQL, PostgreSQL, MongoDB
Visualization	Matplotlib, Seaborn, Tableau, Power BI

Professional Experience:

Pfizer | New York, New York
AI/ML Engineer

May 2025 - Present

Responsibilities:

- Built NLP and predictive analytics models for clinical document analysis, patient risk assessment, and healthcare data processing workflows.
- Developed Generative AI solutions using Large Language Models (LLMs) integrated with Retrieval-Augmented Generation (RAG) frameworks to enable intelligent automation and knowledge retrieval systems.
- Implemented robust MLOps pipelines with CI/CD automation, experiment tracking, model versioning, and continuous monitoring to ensure reproducibility and scalability of ML systems.
- Deployed machine learning models as RESTful APIs using Flask and FastAPI, enabling real-time inference and seamless integration with enterprise applications.
- Conducted exploratory data analysis (EDA) and statistical analysis to uncover insights, detect anomalies, and support data-driven business decisions.
- Engineered and optimized features using advanced techniques such as encoding, normalization, feature selection, and dimensionality reduction to improve model accuracy.
- Worked with large-scale structured and unstructured datasets using Python, SQL, Pandas, NumPy, and distributed data processing frameworks.

- Applied hyperparameter tuning, cross-validation, and ensemble learning methods (Random Forest, XGBoost, Gradient Boosting) to enhance model performance and generalization.
- Developed predictive models for patient outcome analysis and clinical risk forecasting using structured and unstructured healthcare datasets.
- Integrated cloud platforms (AWS, Azure, GCP) for scalable model training, deployment, and monitoring in production environments.
- Collaborated with cross-functional teams including data engineers, software developers, and business stakeholders to deliver AI-driven solutions aligned with organizational goals.
- Implemented model monitoring strategies to detect data drift, concept drift, and performance degradation, ensuring long-term model reliability.
- Improved clinical document retrieval accuracy by 28% using embedding optimization and RAG-based retrieval pipelines.
- Improved model interpretability using explainability techniques such as SHAP and LIME to enhance transparency and trust in AI systems.
- Automated machine learning workflows to streamline development, reduce manual effort, and accelerate deployment cycles.

Environment: Python, SQL, TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, LLMs, RAG, Pandas, NumPy with ML/DL models (CNNs, RNNs, Transformers, XGBoost) and advanced feature engineering techniques.

Exxon Mobil | Irving, Texas
AI/ML Engineer

Jan 2023 - April 2025

Responsibilities:

- Architected and deployed scalable machine learning systems that integrate data ingestion, distributed training, and real-time inference pipelines for production-grade AI applications.
- Developed and fine-tuned large-scale deep learning models using transfer learning and domain adaptation techniques to improve performance on specialized datasets.
- Built enterprise-grade Generative AI applications by integrating Large Language Models (LLMs) with vector databases, embedding pipelines, and Retrieval-Augmented Generation (RAG) frameworks.
- Designed and implemented robust MLOps ecosystems with automated CI/CD pipelines, model validation, monitoring, and governance to ensure reliability and reproducibility.
- Engineered low-latency, high-throughput inference systems optimized using model compression, quantization, and hardware acceleration techniques.
- Implemented advanced ensemble techniques, including stacking and blending, to enhance predictive performance and reduce model variance.
- Developed AI-powered knowledge retrieval systems for engineering and operational teams using transformer models and semantic search techniques.
- Built distributed data processing pipelines using Spark, Databricks, and cloud-native tools to process high-volume refinery and manufacturing telemetry data for predictive modeling and operational analytics.
- Reduced anomaly detection response time by 35% through optimization of real-time streaming inference workflows.
- Engineered real-time AI systems using streaming data architectures to enable continuous predictions and rapid decision-making in time-sensitive applications.
- Implemented distributed training pipelines leveraging multi-GPU and cloud-based clusters to accelerate model training on large datasets.
- Designed anomaly detection and predictive maintenance models for refinery equipment using telemetry data, statistical modeling, and deep learning techniques to improve operational reliability and reduce downtime.
- Implemented model governance and compliance frameworks including versioning, audit trails, bias detection, and explainability mechanisms.
- Developed ML workflows using AWS SageMaker pipelines and Databricks for automated model training, deployment, and monitoring of manufacturing AI systems.
- Built predictive maintenance and operational optimization models using telemetry data and anomaly detection techniques.
- Optimized cloud-based AI infrastructure for performance and cost efficiency using autoscaling, distributed compute, and resource management strategies.
- Developed automated retraining pipelines triggered by data drift and performance degradation to ensure continuous model improvement.
- Built semantic search systems using embeddings and vector similarity search to improve retrieval of refinery and operational documentation.
- Collaborated with stakeholders to define AI strategies, identify high-impact use cases, and deliver scalable solutions aligned with business goals.

Environment: Python, SQL, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LLMs, RAG, Databricks, AWS SageMaker, Spark, PySpark, vector databases, and distributed training (multi-GPU, cloud clusters) for large-scale AI systems.

AgFirst | Columbia, South Carolina
ML Engineer

July 2020 - Dec 2022

Responsibilities:

- Developed and deployed ML models using Python frameworks such as Scikit-learn, TensorFlow, and PyTorch for real-world applications.
- Engineered robust feature pipelines using techniques such as encoding, normalization, dimensionality reduction, and feature selection to improve model performance.
- Performed exploratory data analysis (EDA) to uncover patterns, trends, and anomalies and support data-driven decision-making.

- Optimized model performance using hyperparameter tuning, cross-validation, and ensemble learning techniques including Random Forest, XGBoost, and Gradient Boosting.
- Deployed machine learning models as RESTful APIs using Flask/FastAPI for real-time inference and integration with enterprise systems.
- Built scalable data pipelines using Python, SQL, and distributed frameworks to process large volumes of structured and unstructured data.
- Implemented model monitoring systems to track performance metrics, detect data drift, and ensure model reliability in production.
- Developed time-series forecasting models using ARIMA, LSTM, and Prophet to analyze seasonal financial and lending trends.
- Improved forecasting accuracy by 22% for seasonal financial trend prediction models by optimizing feature engineering workflows and implementing ensemble-based forecasting techniques.
- Developed customer analytics and financial forecasting solutions to support lending and operational decision-making workflows.
- Integrated machine learning workflows with cloud platforms (AWS, Azure, GCP) for scalable training, deployment, and monitoring.
- Applied regularization techniques, model optimization strategies, and performance tuning to reduce overfitting and improve generalization.
- Collaborated with data scientists, data engineers, and business stakeholders to translate requirements into scalable ML solutions.
- Automated machine learning workflows using CI/CD pipelines and orchestration tools to improve development efficiency and reduce deployment time.
- Worked with large-scale datasets and optimized SQL queries for efficient data extraction, transformation, and analysis.

Environment: Python, SQL, Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy with ML techniques (XGBoost, Random Forest, ARIMA, LSTM, Prophet) and data processing frameworks for large-scale analytics.

JM Family Enterprises | Deerfield Beach, Florida
ML Engineer

Oct 2018 - June 2020

Responsibilities:

- Performed in-depth exploratory data analysis (EDA) on large and complex datasets to uncover patterns, trends, correlations, and actionable business insights.
- Built and optimized predictive models for classification, regression, clustering, and time-series forecasting to support data-driven decision-making.
- Applied advanced statistical techniques including hypothesis testing, A/B testing, regression analysis, and probability modeling to validate business assumptions.
- Engineered high-quality features using encoding, scaling, transformation, and dimensionality reduction techniques to enhance model performance and generalization.
- Developed and deployed machine learning models using Python libraries such as Scikit-learn, TensorFlow, and PyTorch for scalable production environments.
- Optimized model performance using hyperparameter tuning, cross-validation, ensemble methods (Random Forest, XGBoost, Gradient Boosting), and regularization strategies.
- Built scalable data pipelines for ingestion, cleaning, transformation, and processing of large structured and unstructured datasets using Python and SQL.
- Developed time-series forecasting models using ARIMA, SARIMA, LSTM, and Prophet for demand forecasting and trend analysis.
- Created interactive dashboards and data visualizations using Tableau, Power BI, Matplotlib, and Seaborn to communicate insights effectively to stakeholders.
- Designed clustering and segmentation models (K-Means, DBSCAN, hierarchical clustering) to analyze customer behavior and improve targeting strategies.
- Deployed machine learning models as RESTful APIs using Flask/FastAPI for real-time inference and integration with enterprise applications.
- Increased customer segmentation accuracy by 18% through optimized clustering and feature engineering strategies.
- Worked with large-scale datasets and optimized SQL queries for efficient data extraction and transformation.
- Collaborated with cross-functional teams including business stakeholders, data engineers, and developers to translate requirements into scalable analytical solutions.
- Monitored model performance in production and implemented retraining pipelines to address data drift and maintain model accuracy over time.
- Built customer recommendation and personalization systems to improve automotive service engagement and customer retention.
- Applied data cleaning and preprocessing techniques to ensure high data quality, consistency, and reliability for modeling.
- Leveraged cloud platforms (AWS, Azure, GCP) for scalable model training, deployment, and data storage.

Environment: Python, SQL, Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy with ML algorithms (XGBoost, Random Forest, ARIMA, SARIMA, LSTM, Prophet, K-Means, DBSCAN) and visualization tools (Matplotlib, Seaborn, Tableau, Power BI).

The Bridge Corp | Hyderabad, India
Data Scientist

Sept 2016 - May 2018

Responsibilities:

- Designed end-to-end data science solutions for customer analytics and business reporting by combining statistical analysis, machine learning, and domain-driven insights.

- Conducted exploratory analysis on customer and operational datasets to identify behavioral trends, performance gaps, and actionable business insights for stakeholders.
- Developed predictive models for customer behavior analysis, sales forecasting, and operational reporting to support business planning and decision-making.
- Applied advanced statistical techniques including hypothesis testing, A/B testing, regression modeling, and probability theory to validate business hypotheses.
- Engineered and optimized features using encoding, scaling, transformation, and dimensionality reduction techniques to improve model accuracy and efficiency.
- Developed machine learning models using Python libraries such as Scikit-learn and TensorFlow for customer analytics, forecasting, and reporting applications.
- Improved model performance through hyperparameter tuning, cross-validation, ensemble learning methods (Random Forest, XGBoost, Gradient Boosting), and regularization.
- Built scalable data pipelines for data ingestion, cleaning, transformation, and processing using Python, SQL, Pandas, and NumPy.
- Collaborated with cross-functional teams including business stakeholders, data engineers, and product teams to deliver data-driven solutions.
- Created interactive dashboards and visualizations using Tableau, Power BI, Matplotlib, and Seaborn to communicate insights effectively to both technical and non-technical audiences.
- Implemented clustering techniques (K-Means, DBSCAN, hierarchical clustering) for customer segmentation and behavioral analysis.
- Built recommendation systems using collaborative filtering, content-based filtering, and hybrid models to improve personalization and engagement.
- Deployed machine learning models as RESTful APIs using Flask/FastAPI for real-time predictions and integration into business systems.
- Worked with large-scale structured and unstructured datasets, optimizing SQL queries for efficient data extraction and transformation.

Environment: Python, SQL, Scikit-learn, TensorFlow, Keras, Pandas, NumPy, SciPy with advanced ML/DL techniques (XGBoost, LightGBM, CNNs, RNNs) and visualization tools (Matplotlib, Seaborn, Tableau, Power BI).

Education:

Anurag Group of Institutions

Bachelor of Technology in Artificial Intelligence

Aug 2012 - May 2016